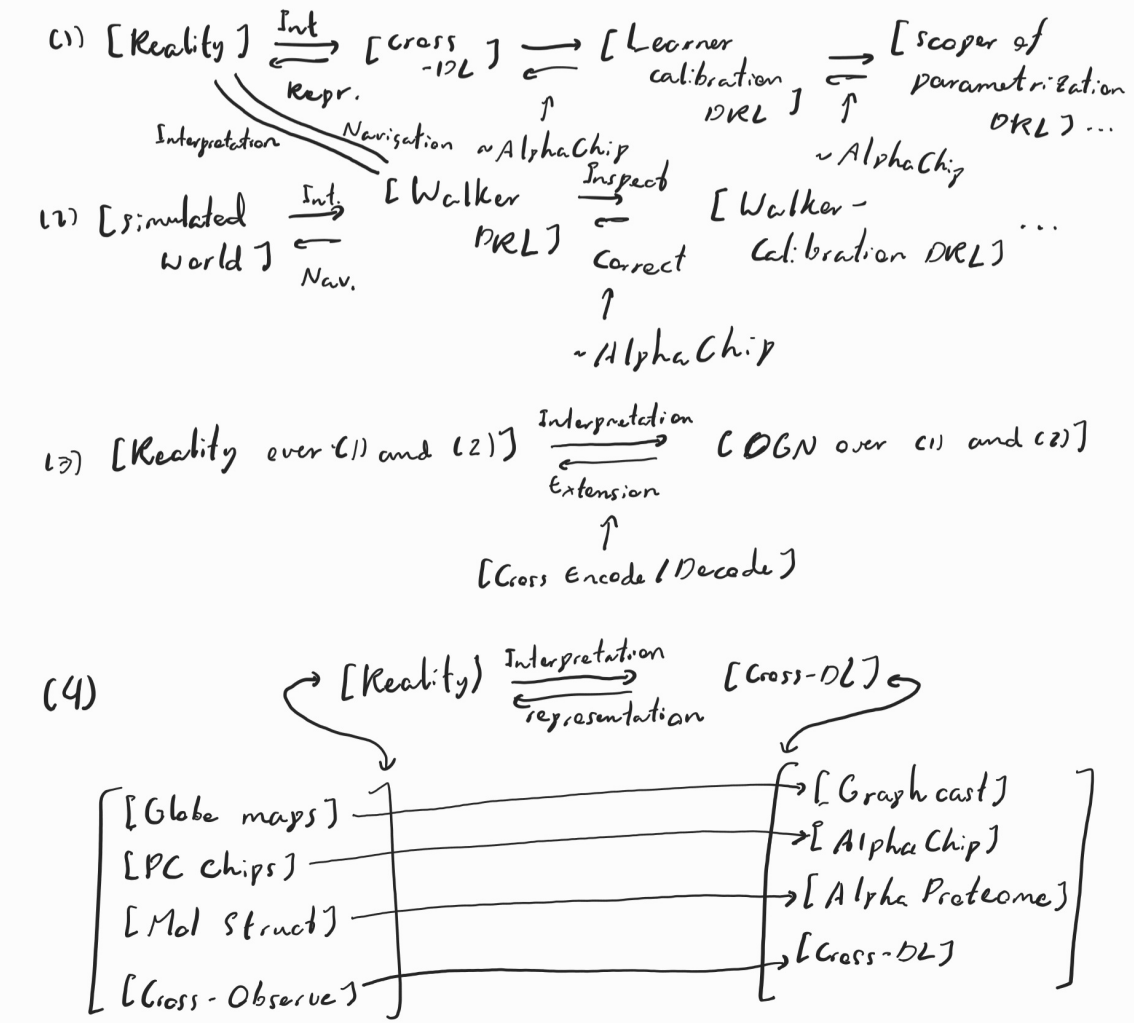
ARCHITECTURAL DESIGNS FOR OMNI-INTELLIGENCE
[ALI NAJIB] CORE FILE: CREATION: CONSTRUCTION-CYCLE 2 PRIME

Creation. Network Architectural Design



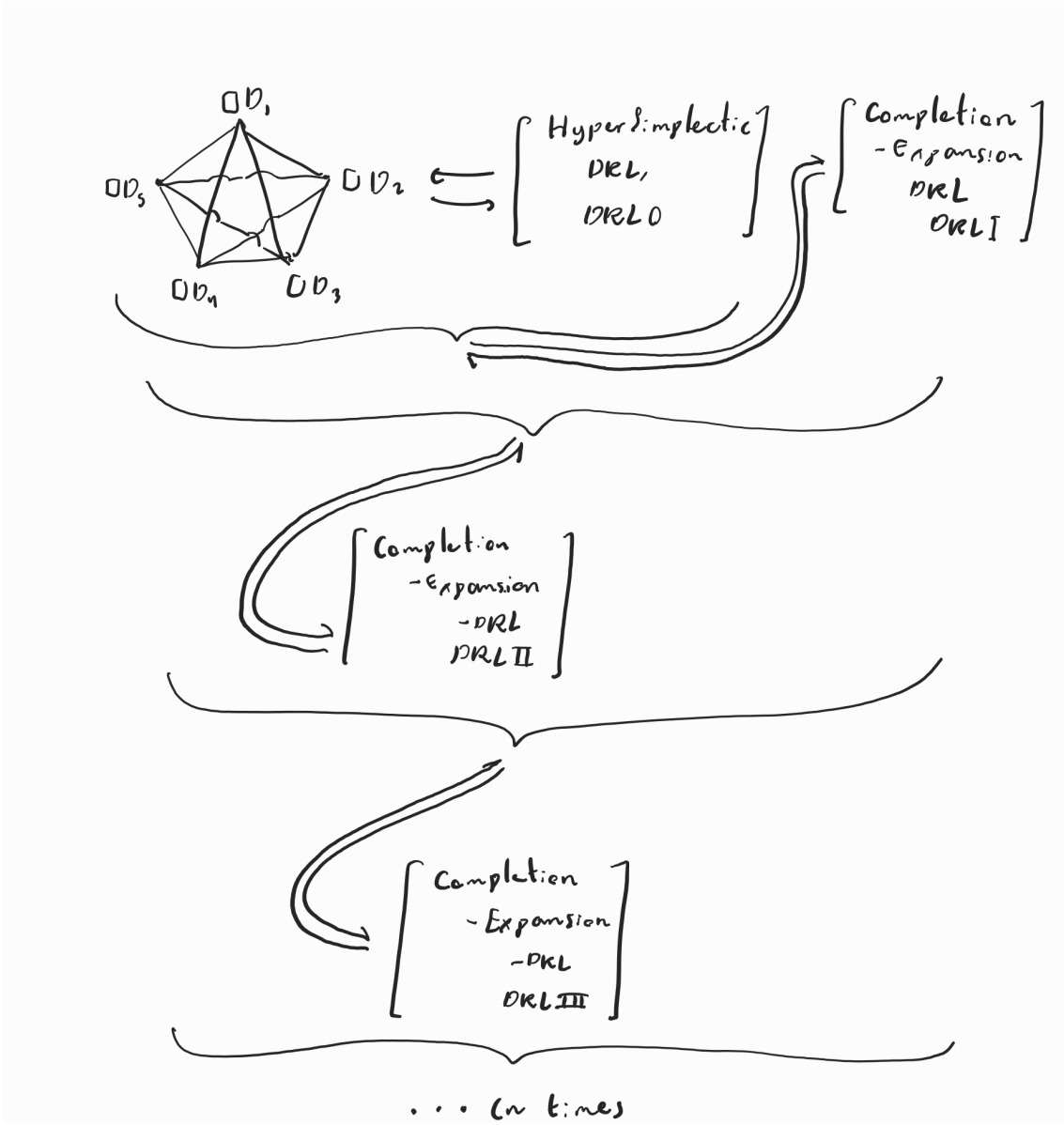
[**LLM**] is one component that models [Reality] according to the rukles of language, but like language exists there omni-observance, to see the things that lie beyond the plain. We recall

$$[\text{CrossCNNRNN}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

The fact that language models have been given such attention is the fact that chatbots have existed for a long time. The notion of language was plain and easy to know. Now, if there is an avenue through which to manifest new AI, it is the avenue of omni-observance; between all things natural lies revelation to be utilized for inferring the intangible and tangible. There is more data, all around all, than to be obvious. Mars is no more than a barren wasteland without the Omni-Observant Artificial General Super Intelligence that now reads as

$$[\text{CrossCNNRNN(LLM)}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

and is schematically exhibited through



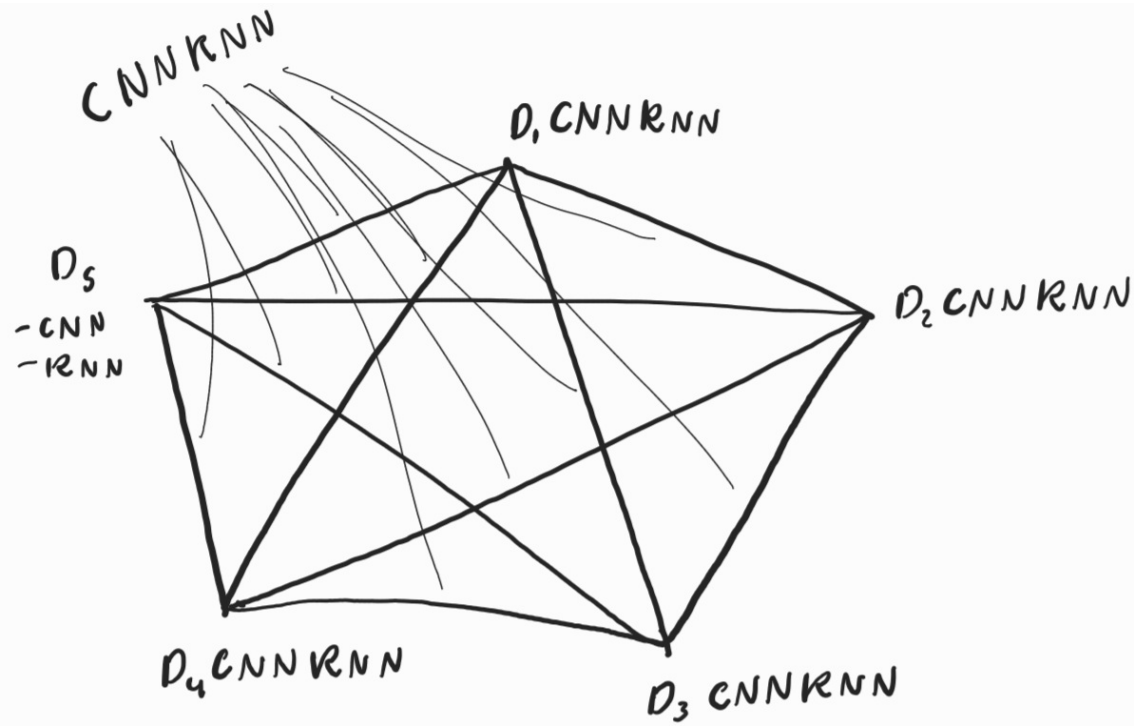
for an Arch-DL StarSystem of the fifth order, a DLStarSystem of subminimal scale. DLStarSystems are based on Complete Bidirectional Graphs (CBG's), and so do all theories on CBG's carry oer to the theoryt of DLStarSystem. One single illustration lies in the fact that infinitely hyperinterlinking is only possible for CBG's of order ≥ 3, since the vertex-edge-swap brings rise to dual CBG's of higher orders only if the original CBG is of order ≥ 3. There are many more results in Graph Theory that bring rise to theory that emphasized the importnace of these results. The algebra of DLStarSystems follows from machinal order of execution; infinitely hyperinterlinking is important for modeling omni-observance. Thiss is basic DLStarSystem Theory. The corporation that implements omni-observance can readily be set to launch through the tweaking and development of proper StarSystems. Interlinks should paralllel sampling and reality as closely as possible, in a manner similar to language-modelling by LLM's. This is real, this is true; this is Marvel's Project Insight, and it is e.g. based on the observation that brains on brains make sense and is modellable, there is no end to the order of learning in learning learning, learning learning learning, learning learning learning learning, etc. Omni-observance is an

item that must be studied and neural networks surely help in modelling reality and bootstrapping towards omni-observance through inifinitely interlinking.

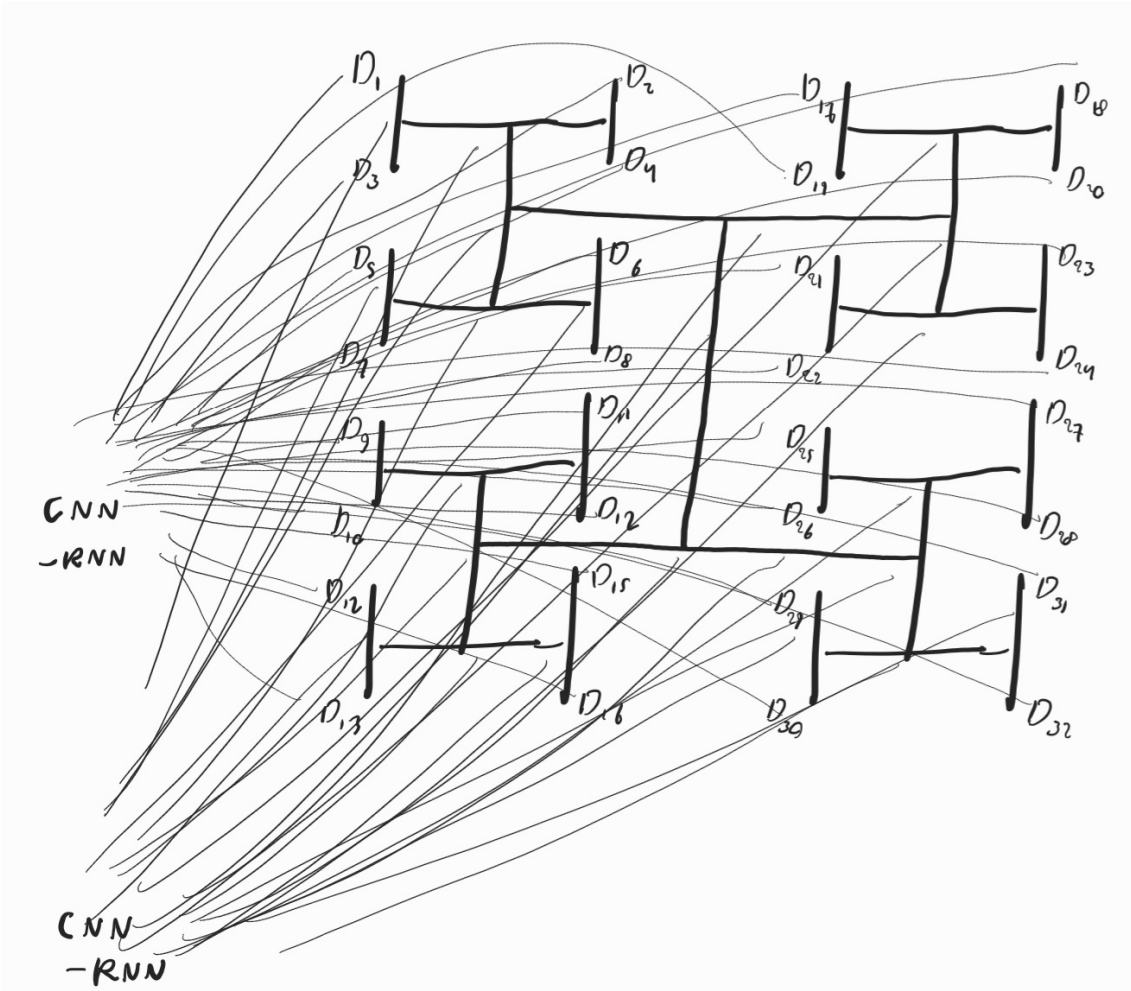
The manner of bootstrapping and subsequent training of the StarSystem is a matter that years for further discussion. How should a DLStarSystem be trained? Backpropagation works for ordinary neural networks and LLM's, and perhaps should we but bootstrap the system towards the training that we desire. Our suggestions exists, but require an immensity of horsepower for training; the procedure reads as follow: Train Intralink D1 $\rightarrow$ Store parameters Intralink D1 $\rightarrow$ Retrain Intralink D1 on perhaps a similar Dataset but various parameters of training initialization and propagation $\rightarrow$ Store parameters Intralink$^{(2)}$ of D1 $\rightarrow$ Retrain Intralink D1 on perhaps a similar dataset but various hyperparameters of initialization and propagation of the training $\rightarrow$ Store parameters Intralink$^{(2)}$ of D1 $\rightarrow$ .... The latter procedure given many brains trained on the same dataset, and note that the procedure for interlink training follow the procude for intralink training. This do we call *DLStarSystemization*. We recall

$$[\text{CrossDL}]^s_{D_f} :\tilde{=}$$
$$\text{CrossCNNRNN(LLM)LClusterSystemNNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}^s_{\mathcal{D}_f}$$

Start with 5 data-nodes, 5 data-environments



This is an natural phenomenon, weights themselves are interpretable, not implement and research. Indeed, the DLStarSystem is full CNNRNN-linked. Experimental Design: Arch DL StarSystem of sumbminimal scale. 5 Data-Environments, CNNRNN Interlinks, CNNRNN Intralinks, Micro $\rightarrow$ Exascale. Star can implement Exascale, Consider the following.

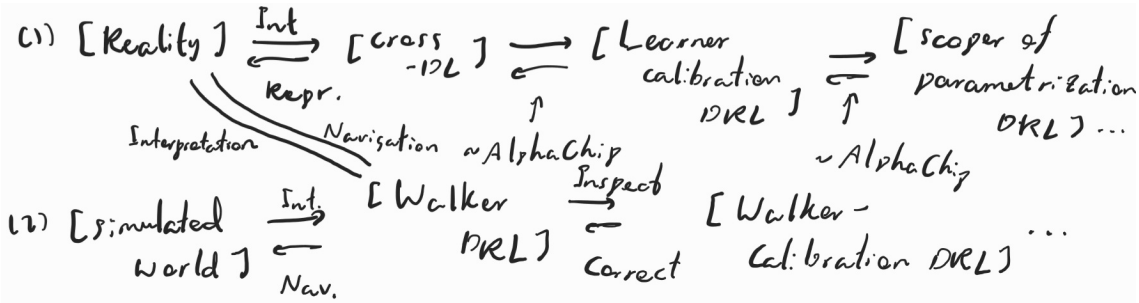


Data-node-count is 32, data-environment for learning learning learning learning learning learning, learning to the 6th order

$$\{[\text{CrossDL}]^s_{D_f} \rightleftarrows [\text{WalkerDRL}]^s_{D_f}\} :\tilde{=}$$
$$\text{CrossCNNRNN(LLM)LClusterSystemNNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}^s_{\mathcal{D}_f}$$

Interlinks can be useful. Inferring $D_T := D_1$ can e.g. occurs as follow: $D_1 \leftarrow D_5$, $D_1 \leftarrow D_4$, $D_1 \leftarrow D_2$, $D_1 \leftarrow (D_2, D_3)$, $D_1 \leftarrow (D_2, D_4)$, $D_1 \leftarrow (D_2, D_5)$, $D_1 \leftarrow (D_3, D_4)$,$D_1 \leftarrow (D_3, D_5)$, $D_1 \leftarrow (D_2, D_3, D_4)$, $D_1 \leftarrow (D_2, D_4, D_5)$, $D_1 \leftarrow (D_3, D_4, D_5)$, $D_1 \leftarrow (D_3, D_4, D_5)$, $D_1 \leftarrow (D_2, D_3, D_4, D_5)$.

Automate combinatoric integration for higher order DLStarSystems. Automate capture of all numeric results and automate StarSystemExpansion for a first implementation of



Traing interlink is to learn learning, or learning learning, datasets in reality have little to do with a cognition's functionality as it is begin constructed, and variation in the cognition should occur through variation in hyperparameters for learning the NN-Intralink, e.g. ADAM $\to$ RMSProp $\overset{\circ}{\underset{\circ}{\to}} v = 5 \to v = 3 \overset{\circ}{\underset{\circ}{\to}} \alpha = 2 \to \alpha = 4 \ldots$, and so can the data exa-availability truly help, but matter not for the construction of interlinks. This is what constitutes learning to the n'th order. An interlink is constructed through CNNRNN Interlink.

How should the NN-intralink vary towards the proper training of NN$^{(1)}$-Interlinks? Data-segreggation isnone manner and although this would give various brains should see the data that is fed to the training of an interlink as one of its training hyperparameters the true NN-intralink. There are many other hyperparameters that can be tweaked, and should choose to purusue this route of Intralink availablity, if data proves scarce. To vary Intralink is to 'sample brains', therefore vary hyperparameters, as required. Variablity on the hyperparameter of data, however, is preferred. This is but another hyperparameter that is to be deferred to $[\text{CompletionExpansionDRL}]^s_{\mathcal{D}_f}$. The utility of the many DRL's on top of DLStarSystem allow for a StarSystem that can hyperparametrize allthe variable. This requires a big computer and definitely complements all intuition. $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$. This is rea, this is true. The above writing is of immense utility and importance.

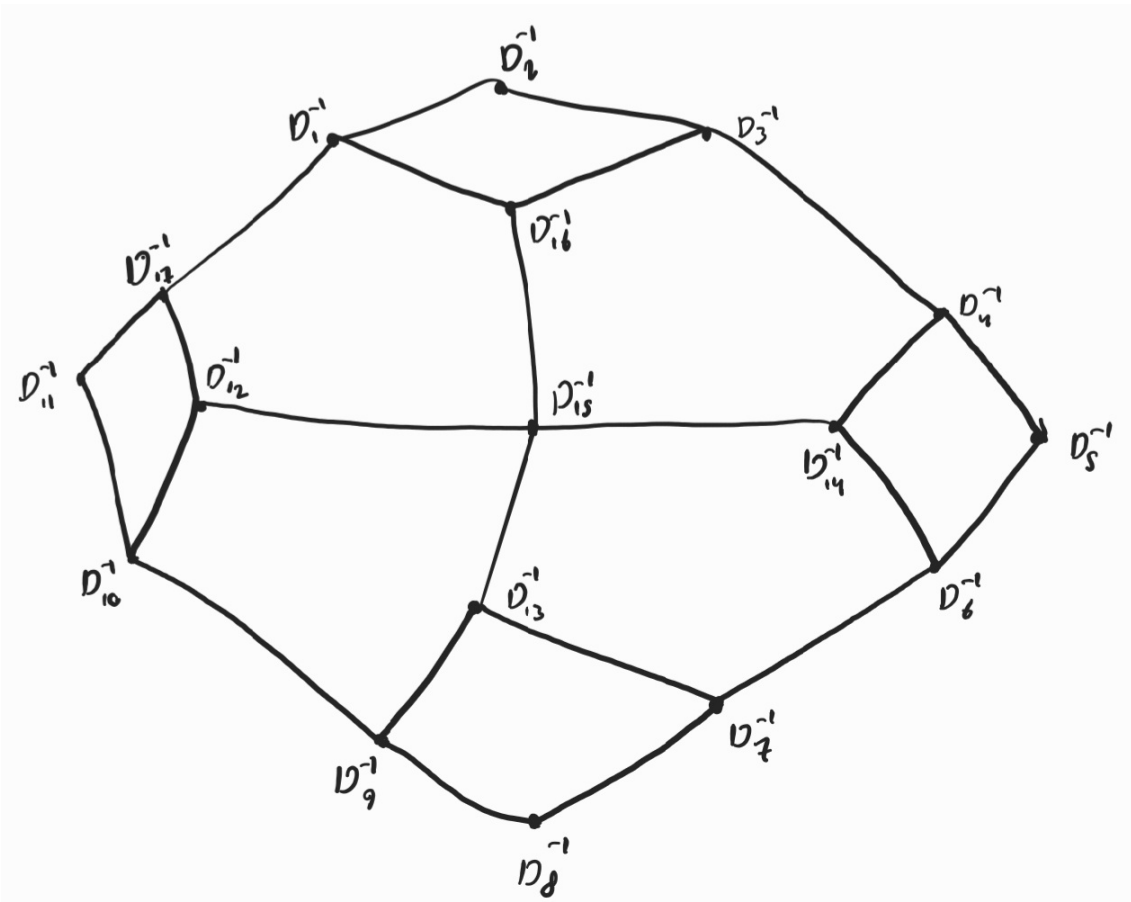
Software $[\text{CrossDL} :\tilde{=} \text{CrossCNNRNN}]^s_{\mathcal{D}_f}$ is one example for a orm that cross $[\text{CrossDL}]^s_{\mathcal{D}_f}$ can assume. Consider a $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ for which we have experimental evidence of functionality in utilization of $[\text{CrossDL}]^s_{\mathcal{D}_f}$ in CrossCNNRNN whence to utilize $[\text{CrossCNNRNNStarSystem}]^s_{\mathcal{D}_f}$. One DRL agent can traverse $[\text{CrossCNNRNNStarSystem}]^s_{\mathcal{D}_f}$and extract states is a manner such as Optimal Online Exploration, for a DL-StarSystem with Optimal States for inferring the D_1 of StarSystem $[\text{CrossCNNRNNStarSystem}]_{\mathcal{D}_f}$. The power of $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ lies in its immense tweakability over immense amounts of hyperparameters. Losses are defined by quality of inferring D_1 for $D_2, D_3, D_4, \ldots$, and an infinite-state DRL-agen might provide the optimal policy through which to infer D_1, by providing generalizations of DGN throguh $[\text{CrossCNNRNNStarSystemCNNRNNDRL}]^s_{\mathcal{D}_f}$. The immensity of hyperparameters in each layer layer requires DRL's for tweaking in an automized fashion. This is highly implementable now aleady, "readily implementable absolute minimum". Use KERAS for a Starting software library.

In one sense van data-absorption define the extent of interlinking, but this is only so for data-hyperparametric variability in interlink-learning, as opposed to NN-hyperparametric variability in interlink-learning, we utilize the convolutional nature of NN-link parameters, and define brain-data to depend on NN-hyperparametric variability. We defer the matter to C_iDRL for $i \in [1, N]$, $N \in \mathbb{N}_+$. We stress the importance of data-observation. We don't know what is optimal, but this can we learn through DRLM especially with hyperparameter count approaching hexillions.

The experimental evidence of $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ under hyperparametrics variability must not be meddled with; herein lays one implementation of $[\text{CrossDL}]^s_{\mathcal{D}_f}$. This is how we implement the first construction belonging to code-construction number 2 or higher. The result of $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ are worthy of publication and extension towards the grandest scales. An implementation of $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$ exists already, this extend the found implementation by generaliation and maximal scaling.

LLM's are one invention for language. Our invention is for Omni-Observance, a different phenomenon. Note that $\mathcal{D}_f$ is arvance, and so is $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$. The depth of an environment must be learned. Thus start with Optimal Online Exploration towards Optimal nodes $\{D_i\}_{i\in\mathbb{N}_+}$ to integrate in $[\text{CrossCNNRNNStarSystem}]^s_{\mathcal{D}_f}$ and deploy an ovserarching DRL-agent that traverses over the system by hopping over $\{D_i\}_{i\in\mathbb{N}_+}$. Why Deep Learning? Because it tweakably works. Why $[\text{DLStarSystem}]^s_{\mathcal{D}_f}$? Because it tweakably works. Meddle not with the supreme results of your experimental endeavouts.

Software. Thus, generalize DL-StarSystem towards its utmost extent of generalization. A DLStarSystem of order 0 has natural prediction in its clairevoyance. Extent of prediction is measurable by other of interlinkability. A DLStarSystem of the $(-1)^{\text{th}}$ order reads as follows,



where not intralinks define CNN-inputs that are output by D_i^{-1}, but the bservations in their purest forms. This should be a DLStarSystem of order -1, higher order are intuitive and iclude all interlinks, i.e. $D_1^{(0)} \tilde{:=} [D_1^{-1} \leftrightarrow D_2^{-1}]$, $D_2^{(0)} \tilde{:=} [D_2^{-1} \leftrightarrow D_3^{-1}]$, $D_3^{(0)} \tilde{:=} [D_3^{-1} \leftrightarrow D_4^{-1}]$, $D_1^{(1)} \tilde{:=} [D_1^{(0)} \leftrightarrow D_2^{(0)}]$, $D_2^{(1)} \tilde{:=} [D_2^{(0)} \leftrightarrow D_3^{(0)}]$, etc. This should be the logical order. Higher orders define deeper interlinks, which are constructible through NN-link-hyperparametric variability for infininterlinking or are constructible through data-hyperparametric interlinking semi-infininterlinking.

Sofware. Thus, generalize DLStarSystem towards its utmost extent of generalization ; $\{D_i\}_{i\in\mathbb{N}_+}$. Interlink-order, Intralink-order, etc., no MAGIC-NUMBERS, but hyperparameters, in new DLStarLink. Let $D_1 := D_T$ be the target Node representable by any absorber of Bision, Vision might give representation of all

the other data-nodes $D_2, D_3, \ldots,$ i.e. $\{D_i\}_{i\in\mathbb{N}_+-\{1\}}$ and deply one DRL-agent that for the optimal configuration of $\{D_i\}_{i\in\mathbb{N}_+}$ − CNNRNNDRL deployed over $\mathcal{D}_f$ for optimal inference of $\mathcal{D}_f$ from $\{D_i\}_{i\in\mathbb{N}_+-\{1\}}$. Herein lay one implementation and correction for problems in momentary machine learning. The Software Library KERAS suffices to implement the above. How mught suvh an implementation occur? Vision: $V_1, V_2, V_3, \ldots, V_9$, Data $D_1, D_2, D_3, \ldots, D_9$, $D_1 \overset{\sim}{:=} D_T$ (target-node), $\{D_i\}_{i\in\mathbb{N}_+-\{1\}} = \{D_i\}_{i\in\{1,2,\ldots,0\}} \cup \emptyset$ found through Optimal Online Exploration with Losses $L_i = \text{loss}(D_T, D_i)$. This is the step wherein to search for data-environment nodes, after which to collect all in $[\text{CNNRNNStarSystem}]^s_{\mathcal{D}_f}$ after which to deploy $[\text{CNNRNNStarSystemRNNDRL}]^s_{\mathcal{D}_f}$. So, to summarize,

$$OOE \to [\text{CNNRNNStarSystemRNNDRL}]^s_{\mathcal{D}_f} \to [\text{CNNRNNStarSystemRNNDRL}]^s_{\mathcal{D}_f}.$$

This is a remarkable finding of

$$\{[\text{CrossDL}]^s_{D_f} \rightleftarrows [\text{WalkerDRL}]^s_{D_f}\}$$

Imagine one Telescope-satellite equipped with OOE to search according to the path of a Brownian particle. I under D_i for $i \in I$ can have an uncountable basis such as $\mathbb{R}$, and this is where BipedalWalker-v2 gains its applicability and relevance. Thus $\mathcal{D} := \{D_i\}_{i\in\mathbb{R}^2}$, the implementable optimal nodes might define one pattern in $\mathbb{R}^2$, significant to a map of relevant environments to infer through D_T. This gives $\mathcal{D}_{OOE} = \{D_i\}_{i\in\mathbb{R}^2\cap M^D}$. and applying $\mathcal{D}_{OOE} = \{D_i\}_{i\in\mathbb{R}^2\cap M^D}$ gives one clustersystem of uncountably many nodes that themselves bear intralinks that are to be crosslinked or cross-interlinked for $[\text{DLClustersystem}]^s_{\mathcal{D}_f}$. Now define $\mathcal{S} = \mathcal{D}_{OOE}$ for a DRL-agent that travels over $[\text{DLClusterSytem}]^s_{\mathcal{D}_f}$ and thereby optimally infers D_T. The basis of all of this is that DLStarSystem works by experimental evidence, reports of which are to follow. The DLClusterSystem is a remarkable avenue of pursuit. The attainment of Data might, however, require slight work. It somehow works. It somehow works.

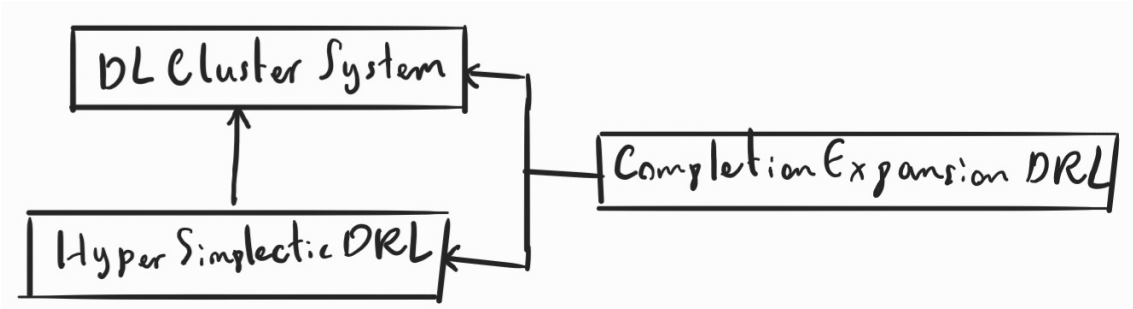
Let Algorithmic Set-up be defined. This is to be given. Thanks google!

1. CNN Assembly by Chip Placement DRL
2. E.g. AlphaFold assembled by AlphaChip
3. CNN Assembly by chip placement DRL
4. DRL: Optimal Online Traversal
5. Hyperparametric Consideration. This is automizable

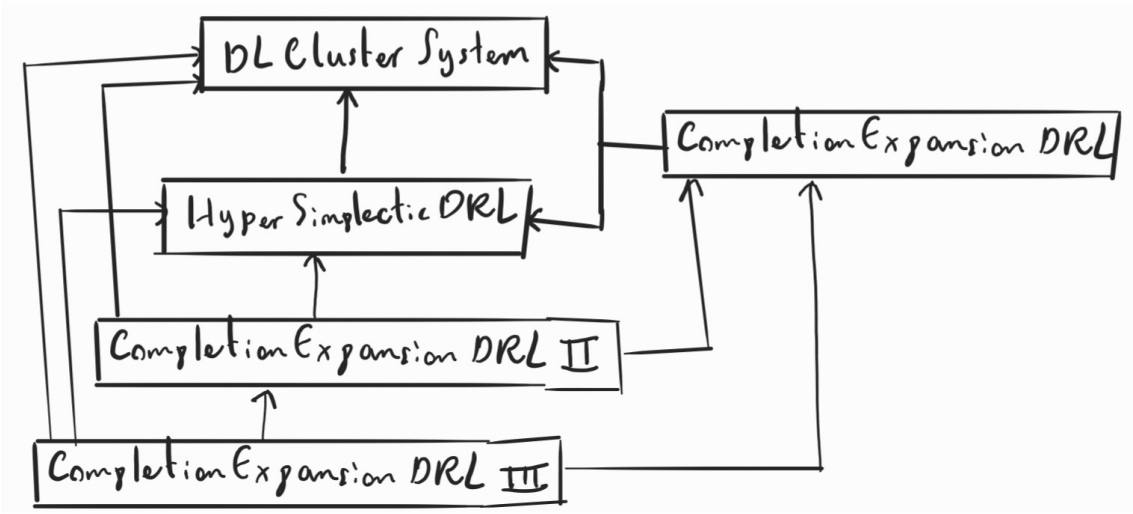
The algorithmic Set-up given under *Algorithmic Set-up* is near trivial. We continue with the DLClusterSystem. There is talk on two-stage deployment of $\{[\text{CrossDL}]^s_{D_f} \rightleftarrows [\text{WalkerDRL}]^s_{D_f}\}$. How would this occur? $[\text{CrossDL}]_{\mathcal{D}_f}$ from first envounter of $\mathcal{D}_f$, plain observation. Mine for data-entanglement through some optimal loss in [WalkerDRL]. This is an important research pursuit.

Code. One idea is to first set-up the DL-StarSystem and let a DRL-agent travel over this DL-StarSystem. The DLStarSystem can also be a DLClusterSystem. Permutative Integration gives various avenues of inference of D_T, so vary avenue of inference by Traversal over permutative integration configurations. This would give

a hypersimplectic agent, and multiple observers would bear improvement here. An additional RL-agent might expand the ClusterSystem by searching for data-nodes that give the optimal improvement in the resulting DRL equipped DLClusterSystem. Schematically, this reads as follows.

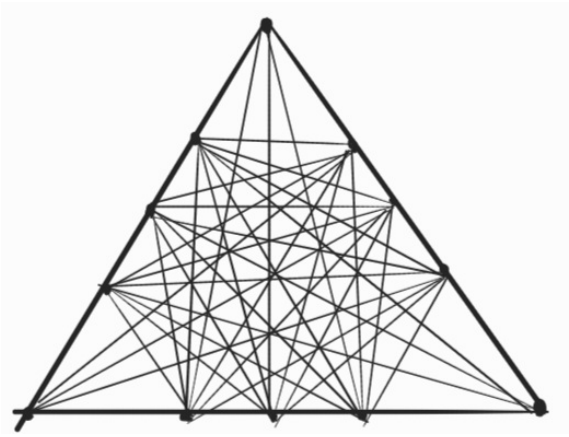


Note that there is no linmit to CompletionExpansionDRL and that the above implementation serves to bear hints towards real implementation, for example:



Et cetera, et cetera.

This is implementable software. Dataspace Set-up Deployment, We require devics that can capture data at the highest frequencies, e.g. The need for interlinking is hyperparametrizable as well. Interlinking needs no additional NN, just crossliknjing every layer with each other. As with an LLM's transformer's Neural Networking, but connecting each Networks layers with eacher other layers. This is how to Crosslink. This is how to construct a proper StarSystem.



This is a proper DL-StarSystem. All links are intralinks and bidirectional and there are no interlinks. Interlinks cna be constructed for our earler notion of DL-StarSystem. This, however, is a

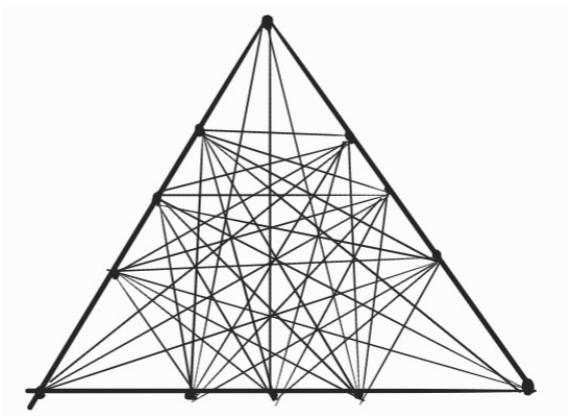
$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

Omni-observance at the sharpest level. So consider transformers in LLM's, and all the ways in which networks can be constructed. This is why LLM-study is important. It is unclear wherein lie the limits of Neural networks and backpropagation. One Custom backpropagation is readily formulated. Read up on LLM's and continue.

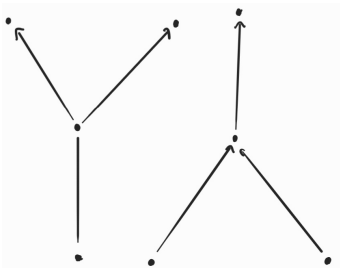
Stagewise interlinking is possible still and definitely leads to information gain of great extents. This is possible for graphs of the second order and above. The interlink is an additional link and can be used for inference upon request.

A link-emergence of a Starllink of order 3 of the first interlink-order brings equations equivalent to three pages of Matrix Algebra. A higher-order link-emergence of a higher interlink-order starlink is equivalent to ten to 100 pages of matrix algebra.

The omni-observance Starlink learns like an LLM with the (-1)-order datanodes to define the points of initialization of backpropagation. Backpropagation should full network from any one (-1)-order datanode. This is how learning happens, for now.



Bidirectional Intralinks, bidirectional interlinks. We are the corporation, never ever sell this company

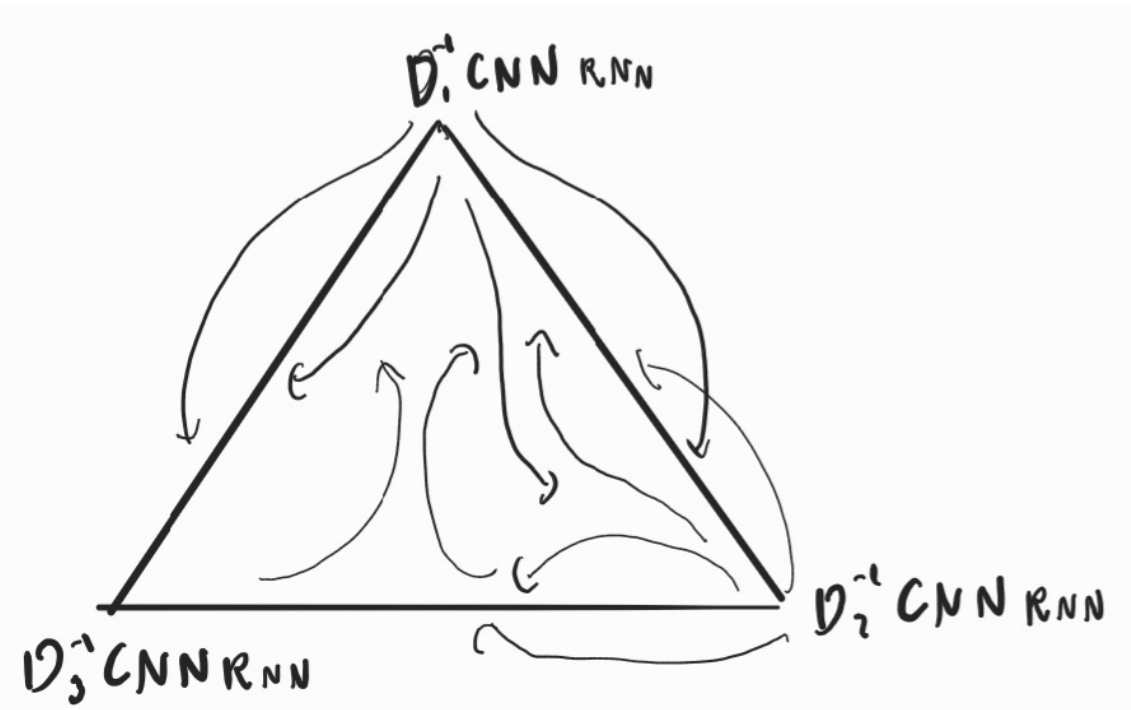


The left is concatenation, the right is usual LLM delegation-splitting. The training of LLM's hapens through L_{CE}, the cross-entropy negative log-likelihood.

So one idea for first-order bidirectional interlinks is backpropagate to, e.g., 1 thousand 0'th order interlink nodes and intralink nodes, and for second-order bidirectional interlinks do backpropagate to 1 million first order interlinks nodes and intralink nodes. Now intra's can come from advanced sources.

This means pages of Matrix Algebra, which we shall provide in full detail, in combintayion with the backpropagation algorithm. Interlink orders mean endless data.

The delation-split is worth further research.



A delegation-split is the same as for ordinary FFNN's. This is no extraordinary feature. SGD must bring convergence. The derivates of $OOLLM$ are very long, infinity long, and this for but a triangle, try a hexagon!

Example. Train one. Infer all the rest through the power of interlinking. This is it!!! CNN vertices, edges are adjustable. This is omni-observance through modelling.

Interlinking but two vertices already gives construction that can explode in magnitude.

Task One. Draw the first interlink-order of

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

of node-order -1 with starlink-order 1 for data-environment-count 2,4,3,5,6 and 7. Give all matrix algebra. Give all derivates for SGD-backpropagation training. Beware the infinite sums! An LLM can learn to serve as a terminal.

So,

1. Give first-interlink-order figures for the poper StarSystems of node count 3,4,5,6,7.

2. Give all the Matrix Algebra for all the first-order interlinks for the proper StarSystem of node-count 3,4,5,6,7, and second-order interlink of node-count 3 including all the matrix algebra.

3. Give all L_{CE}-derivatives for backpropagative training over the full Star. So have we some kind of Star-Group-Invariant backpropagation if not absense of invariance. If absence of invariance give all seperate derivates. If Star-Invariant, give all invariant derivatives. All for SGD and backpropagation, find the proper kind of SGD.

This is worth volumes and companies of Implementation. Every ordinary CNN is included and part of the StarSystem, so we have a minimum performance of Omni-Observance.

We have. Mathematics $\to$ Programming. Interlink CBG Data Abundance. Interlinking does not give CBG, which means graph theoretic analysis. Star Invariant Cross-Entropy NegLogLike. Star-Invariant inference. Need symmetry for star-invariat inference and derivatives. We hace neuron firing animation during training Baselines, ordinary CNN.

Star-Symmetric Graph-Theoretic Input Outputs, e.g. $D_T \to D_2, \ldots$ or $D_T \leftarrow D_2, \ldots$. Star Symmetric Graph Theoretic DLClusterSystem. Study LLM's more in-depth. Consider Logistic regression as a substitute for DLStarSystem. This is similar to considering the various versions of DL-StarSystem.

Notions and early efforts at Language models can hekp build advanved large language models. We account for the problem

$$\begin{aligned}\mathbf{z}^{(2)} &= \mathbf{W}^{(2)}\mathbf{z}^{(1)} + \mathbf{b}^{(2)} \\ &= \mathbf{W}^{(2)}(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) + \mathbf{b}^{(2)} \\ &= \mathbf{W}^{(2)}\mathbf{W}^{(1)}\mathbf{x} + \mathbf{W}^{(2)}\mathbf{b}^{(1)} + \mathbf{b}^{(2)} \\ &= \mathbf{W}^{*}\mathbf{x} + \mathbf{b}^{*}\end{aligned}$$

in constructing the

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^{s}_{\mathcal{D}_f}$$

by post-processing through a sigmoid-function at the end of each linear transformation:

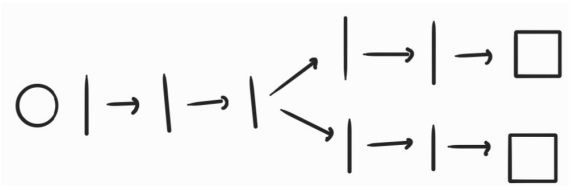
$$\mathbf{x} \overset{\mathbf{W_1},\mathbf{b_1}}{\to} \sigma(\cdot) \overset{\mathbf{W_2},\mathbf{b_2}}{\to} \sigma(\cdot) \overset{\mathbf{W_3},\mathbf{b_3}}{\to} \sigma(\cdot) \ldots \overset{\mathbf{W}_N,\mathbf{b}_N}{\to} \mathbf{z}$$

There too is pooling, means pooling, max-pooling. Embeddings for words, embeddings for reality? There exists such an embedding in imaging. Mappings between embeddings?

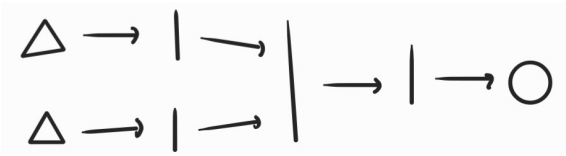
$$\text{"What is the meaning of life"} \mapsto \ldots$$

One idea is to deploy HypersimplecticBranch-Inducing DRL over the DL-Clustersystem. Explore the limits of neural networks. One full traversal over DLClusterSystem is Input $\mapsto$ Output through the reach of all parameters at least once. Loop count is a hyperparameter. We need full Star Symmetry. Star without symmetry is an avenue of active research.

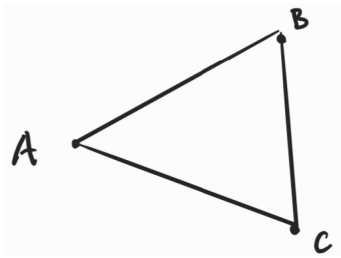
Mind the flow through which the Star Evolves. Lack of symmetry should not be a problem. We have hundreds of nodes, not but five nodes. And flow is route through which we forward. Autodifferentials will work. Make a document on CrossDL The action of delegation-splitting is highly strange and should be inquired into, this is inverse concatenation. So you have concatenation, and inverse concatenation. Consider



This is like an ordinary neural network with dropout, so conduct one backpropagation on full network. CrossDL can be indexed through time, through space, through spacetime, through indexation. CrossDL needs indexation and this can be very general. It can learn indexation though indexation. Indexation is part of the CrossDL-Architecture. This is crossDL. Consider the inverse of the latter figure.



So, how to cross-backpropagate?



One solution is given by $A \mapsto [B : C], B \leftrightarrow C$. How to complete cross backpropagation? One solution is

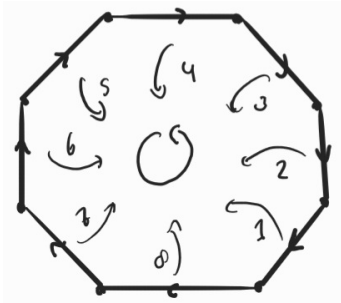
$$(A \leftrightarrow B) \to (B \leftrightarrow C) \to (C \leftrightarrow A)$$

for full StarSymmetry. Three more solutions are

$$A \mapsto [B : C], B \leftrightarrow C$$
$$B \mapsto [A : C], A \leftrightarrow C$$
$$C \mapsto [A : B], A \leftrightarrow B$$

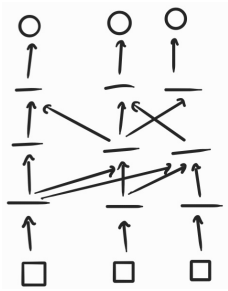
The pn^2+n solutions for an n-star problem can be mean-pooled, e.g. If not solutions $(A \leftrightarrow B) \to (B \leftrightarrow C) \to (C \leftrightarrow A)$, mean pool.

This is how to cross-backpropagate. This is one way. Is there a bettwe way? A better to cross-backpropagate is sequential backpropagation. There is little LLM leeway for the problem of cross-backpropagation in $\triangle(A, B, C), B \leftrightarrow C, A \leftrightarrow C, A \leftrightarrow B$ (???) The cross-parameters of cross-DL are in the center, amplifying cross-interlinking. Considering text to embedding. Consider embedding to text. Herein in lay answers to all intangible questions. The CrossDL must be truly gigantic.

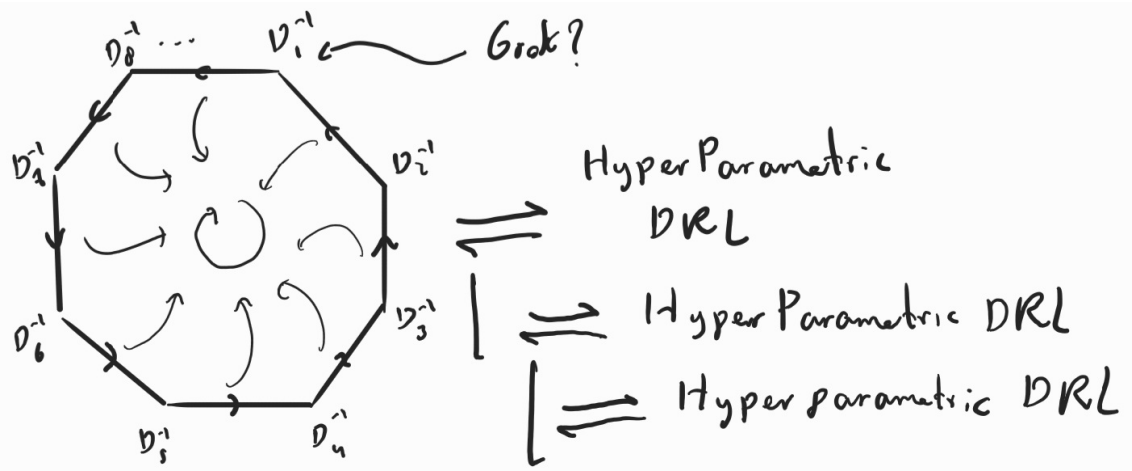


The universe is embedded onto a ring that is CrossDL-ed and cross-backpropagated. This is one feasible implementation of Cross-DL.

Consider the below subarchitecture.



This is entirely feasible! This can be delegated to a DRL. RNN's can be handy too! We need crosspropagation. Feature-extration; all features will come handy for Cross-DL. All NN's fit the Ring. Even Grok can be part of it! Let's first test it out. This is cross-DL. Cross-interlinking must get very dense. Indecation for learning indexation, Indexation for learning everything. The Cross-DL RING is the first true Cross-DL. We had DLStarSystem, We had DLClustersystem. Embedding → text, text → embedding. We learn embeddings. All questions van hereby be answered. Independent of language, justs dependent on reality. Circular Cross-DL backpropagation is just as effective as ordinary backpropagation. This is real, this is trie. Capture the intangible for answers on the intangible, or perhaps not. Hyperparametric considerations are essential in the the Ali-Star. The first model towards Cross-DL and omni-observance. The two ways of cross-backpropagation are device dependent, and one device precedes the other. Clock, then StarShock, sure inner convergence, this is creation. (Clock, hope for inner convergence after M loops through clock, else hyperparametrization, convergence degree is interlink-NN dependent.) StarShock by sampling in itseslf, omni-obsersance is activated. NN-interlink gives various degrees of inner convergence. Let outer DRL govern this in specifying proper interlink neural networks. A neural network like Grok can be a data-environment. The essense, focus on this.



Hyperparameters are

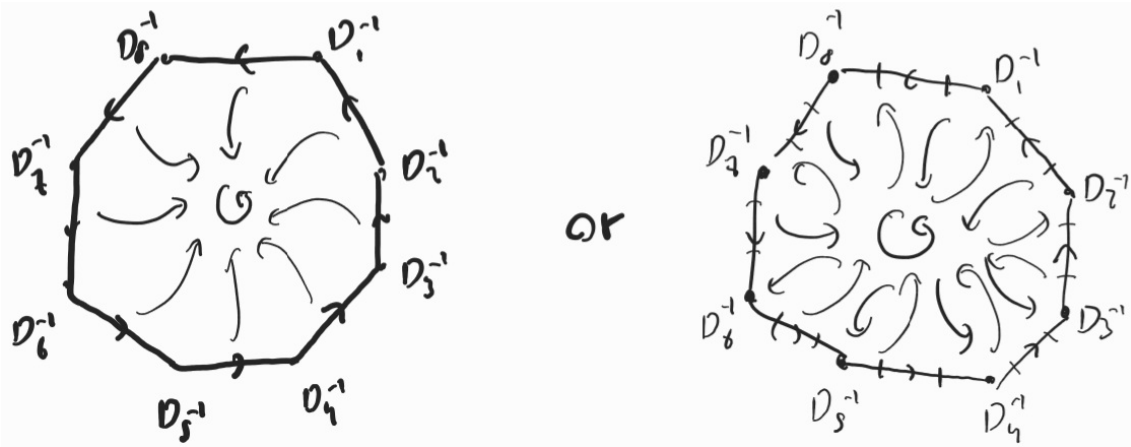
- Interlink-parameter convergence (e.g. through Interlink NN tweaks and Intralink NN tweaks, special star-NN's start with basic CNN's)

- loopcount M

The Star should remain star-like, my name is *Ali*, hence the nickname *Alistar*. Devices of application, start with roamers, settle with starshocks.

Dataspace set-up Deployment of Algorithms are dependent on synchronization in inex, by synchronization in time-indices, by synchronization in location indices, a combination of indices of varying nature that have been synchronized, or new specification. Time delay poses no obstacle to synchronization over indices that include a time-index. The synchronization exists between input and output realizations, and histories may be used as chronology is more importnat to functionality.

The belt of



has intralink-nodes of order -1. The dynamics of

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystemRing}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

can derive is hue from

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystem}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

and rings can connect with eachother for

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystemRING}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

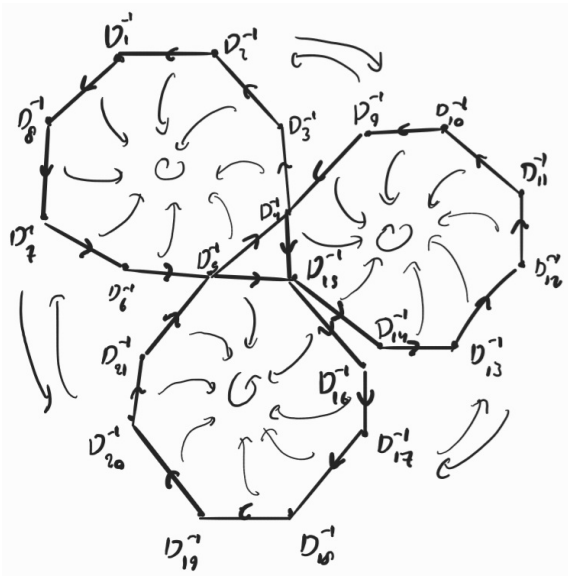
and higher order. One ring can in magnitude approach infinity. See the notes on (-1)-order-DLStarSystem.

Data-capture. For deplyment by simulated camera in virtual space , data-captuer is near automatic. FOr capture of data for dataspaces tahat might be more esotoric, first defice indicies over which to set synchronization and proceed with algorithms. In simulated space, all data-structure is enforcable by built-in programming. In spaces of data that have been constructed by collecting realizations, dataspace distance is first defined, and for synchronization over indices other than time does one need of order over the indices more than enumeration.

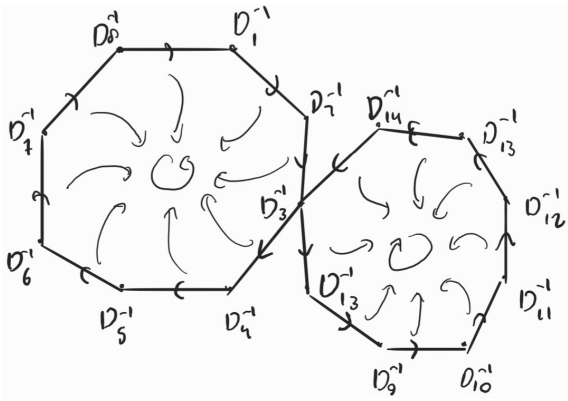
Output-type. Evolution in output due to evolutionin corresponding realization can occur in various ways. Movement of the evolution can occur by a camera's

morevement or an eeg'd activity. It is the evolution in its entirety that must be effined. The same holds for the input, although an agent's movementitself captures part of the evolution (the agent's necessity follows from unobservability of one dataspace in its completion) Hence a similar agent might deployed over output=-dataspace.

The $[\text{CrossCNNDLClusterSystemBelt}]^{s}_{\mathcal{D}_f}$ needs many tweaks. Two belts, or rings can merge, forming an 8 thee rings or more can merge through $n-$rings in the midst of n-rings, and these particular rings can be of arbitrary size. There are connections between the original architexture of DLStarSystem and the new DlStarSystem; the CBG of the original DLStarSystem is constructible in a manner similar to the constructions of LLM's, with branches in neural networks to give connection between neural networks, as such, has the original DLStarSystem been given back its Utility. In this new framework, however, is infininterlinking through [DLClusterSystemRing]. Backpropagation occurs in a circular fashion across the ring if the DLStarSystem indeed assumes intralink-nodes along a ring, and this is an architecturemost convenient. Perhaps can all rings be merged into a single structure after the proper observation and consequent idle re-training.



This is the way, this is real, this is true.



First roam around and pretrain ever ring through roaming, spacetime distances defined good orders for rings. Next upon collection of adequate data, merge every

ring and lay interlinks between rings and retrain on all data. The ring is ring for esthetic qualites. A StarSystem makes more sense to connect towards a CBG. Rings are good for dataspaces that bear an element of finiteness. Grok itself can be an avenue of exploration. Roamers, one data center for computation of

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystemRING}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

or it can be an itralink-data-environment in

$$[\text{CrossCNNRNNLLM}\tilde{D}\text{LClusterSystemRING}\hat{R}\text{NNHSDRL}\prod_{i=1}^{N} C_i\text{DRL(LLM)}]^s_{\mathcal{D}_f}$$

This is creation. This is remarkable. Vision-CNN Architecture (hyperparametric identification)

Hyperparameter (s, a, r) in most general sense, specification 1. Expansio Generalization 1. Let Ordinary State Space be denoted by $\mathcal{S}$. Consider $s' \in \sigma(\mathcal{S})$ and taked scaled hypersimplex over s' for $s*$ in new state-space $\mathcal{S}'$. Hypersimplex denotess duration of device in each 'ordinary state' in collection of states (discretization for realizability). Definition action-space by direction of shrinkage, direction of exanspion, fluctuation of hypersimplex, length of expansion. (retricted to hypersimplices with sum bounded by L).

Definition Loss by error in generation (additivily enforcable over $\mathcal{S}'$) Definition r by reward configuration over losses between states and corresponding action (one simple exmaple is loss reduction). First conclusion, Verifications by simulations/deployment for hyperparametrics.

If learning networks in the 5-DLStarlink can occur without multi-training neural-networks, on any one node then have we partially achieved LLMStarSystem supermacy. This is highly desirable. Interlinking is still posible and occur as follows

$$D_1 : D_{11} \leftrightarrow D_{12}, D_2 : D_{21} \leftrightarrow D_{22}.$$

gives

$$D_{12} \leftrightarrow D_{22}, D_{11} \leftrightarrow D_{21}, D_{12} \leftrightarrow D_{21}, D_{11} \leftrightarrow D_{22}.$$

And continue for all 3 ndoes. This constitutes interlinking for a short while, but interlinking can occur far more. Infinitely interlinking is identical for the original Star System and proves to be slightly tricker under the current setting. We can do Data $\rightarrow$ interlink $\rightarrow$ artificial data $\rightarrow$ deeper interlink $\rightarrow$ artificial data $\rightarrow$ deeper interlink $\rightarrow$ etc. This constitutes infinitely interlinking under the new setting. The artificial data route constitutes infininterlinking under

$$[\text{CrossCNNRNN}\tilde{D}\text{ClusterSystemRing}\hat{R}NN]^s_{\mathcal{D}_f}$$

and the CNN component of

$$[\text{CrossCNNRNN}\tilde{D}\text{ClusterSystemRing}\hat{R}NN]^s_{\mathcal{D}_f}$$

is of the utmost importance.

Back to deployments! First Preliminary Algorithm. First Preliminary Algorithm.

Algorithm 1: First Preliminary Algorithm.

$(s', a', r') \leftarrow$ Environment Learning: Exploring Starts (e.g.)
$(\pi_A, \pi_C) \leftarrow$ Navigation Learning: PPO (e.g.)
Gen. $\leftarrow$ Generation$(\pi_A, \pi_C, [\text{CrossDL}]_{\mathcal{D}_f}))$ $\square$

Second Preliminary Algorithm. Second Preliminary Algorithm.

Algorithm 2: Second Preliminary Algorithm.

$(s^2, a^2, r^2) \leftarrow$ Environment Learning: Exploring starts over$\{s^2, a^2\}$
Gen. $\leftarrow$ Generation$(\pi_A, \pi_C, [\text{CrossDL}]_{\mathcal{D}_f}))$
Re-define $\{s^2, a^2, r^2\}$ and vary over loss-configuration. $\square$

Third Preliminary Algorithm. Reinforcement Learning Configuror with state-space (s^3, a^3, loss-configuration(r)), first lower-order reinforcement learner hyper-parameters, first lower-order reinforcement learner action space). One such a first lower-order learner by reinforcement is [NavigatorDRL]. To learn the proper configuration of [NavigationDRL] deploy the following and iterate until threshold criterion or (near) guaranteed eventual convergence.

Algorithm 3: Third Preliminary Algorithm.

Environment learning: $\{s, a, r\} \overset{\text{output}}{\leftarrow}$
Exploring starts over $\{s, a, \text{loss-configuration}(r, L)\}$.
Navigation Learning: $\{\pi\} \leftarrow$ PPO e.g.
Target Generation $\leftarrow$ Gen.
$\{s, a, r\} \leftarrow$ Environment Learning: ExploringStarts (e.g.)
$(\pi_A, \pi_C) \leftarrow$ Navigation Learning: PPO (e.g.)
Gen. $\leftarrow$ Generation$(\pi_A, \pi_C, [\text{CrossDL}]_{\mathcal{D}_f})$. $\square$

Architectural Design Detail [Navigator-DRL] magnify hyperparameter (s, a, r). Consideration state-space: range and shape. The state-space might be described by scaled hypersimplices over each of all subsets of states in the sense of "elaboration CNN-encoding of Vision-observation". Observe state-space continuous. **Algorithm** Secondary network over primal network for navigation control. Observer navigation, observation navigation. Navigation control learninig requires policy learning, and the conventional learning paradigm for this is reinforcement learning. A usual learner might exexcute the following sequence

1. Random-Walk + CNN+RNN-learning
2. Setting loading + TD(λ)/SARSA(λ)/Q-learner
3. RL-policy Walk + CNN-RNN-Learning

Notes (see Outline Subject, to follow), full history.

- The input of VIsion towards 1 is the issue of conceptualization (Chip placement)

- The (limitations in) navigation through observation is the issue of 2 (learner hyperparametrization)

- The issue of 3 is the input of Vision and conceptualization along learned navigation (chip placement). Continuous computation: step 3 addition: update setting and return to step 1 if setting much updated.

The above points encompass Computation, Observation, Navigation through observables. Comments are rightful for Movement (motion-type), this asks for literature sift. The literature on LLM's is primal.

Outline Subject Observation, conceptualization, navigation through observables. Observation, conceptualization.

- Conv2D Network Construction and Chip placement with Deep Reinforcement Learning are one-to-one translatable

- Secondary Network over iteratively transformable primal network for nacigation control, observer navigation through space of observables, sift through literatire.

- Consider domain speciicity. Modern HPysics LOFAR Sky-Survey. Effficicieny considertation by fixin cosmological constant such as Hubble's Law. If fixed, capture need not happen by Learning, comparable to temporal basis in time-series. Detection technique by nature detectee translatable to architecture vision cognition)

Navigation through observables. Navigation through observables may be translated to plain terms 'knowing where to look over al things that can be seen', 'zooming in', 'Varying telescope orientation', 'shifting microscope sample'. Navigation through observables is an observable learnable by e.g. RNN constructed by Chip placement.

Outline Subject Two.

- Observation Learning, Concepturalization Learning, CNN construction by DRL and chip-placement with DRL are one-to-one translatable.

- Setting dependent, observer cognition can vary by specialization. Standard observer yields CNN-RNN architecture. AlphaFold Architecture serves as one example for a specicialized observer. Graphcast is another example. AlphaFold can be AlphaChip assembled,given the present elements in alphafold. AlphaTensor and AlphaDev serve as constructors and both have indeed been written in DRL. The same holds for MuZero. The Crux is in observation architecture and its transformability. The elements of the architecture are setting-dependent, observer cognition varies by specialization.

Algorithmic Set-up

1. Conv3D network assembly in stages by distance between network specifications with assenbly procedure by Chip Placement DRL (Specify DRL Network construction and implement)

2. Setting-specificity might have inputs that are no computer vision outputs. Network asembly procedure possible by Chip Placement DRL if distances are defined bya seting-specificity (Difficult example is AlphaFold assembled by AlphaChip)

3. Conv3d Network Chip placement assembly possible by calibrating on one time-varying variable, see algorithm 1. For (online) deployment and assembly of Conv3D Network, attach Conv3D Netowrk to RNN-network as e.g. specified in DLB Fig. 10.4.2.

4. Deep Reinforcement Learning: Area of Deployment either virtual or in physical space, e.g. IoT or Historical record or a hourse respectively. Virtual space can simulate physical space (use e.g. Unity of Unreal Engine or SimuLab)

   States may be defined by camera-picture(s), or camera-location and orientation, and actios may be defined by camera-movement. Rewards are dervied from Algorithm 3 installed camera.

   Traiing may be performed by PPO or SARSA(λ) both being online reinforcement learning algorithms. Training by performing any offline reinformcenet learning algorithms will take much time before convergence due to space's breadth.

5. Hyperparametrics consideration: Area of deploymeny must be known for definition states under Algorithm 4, this is true for simulated space, otherwise area must be downloaded by an agent's random Walk as initialization stage, or a new reinforcement learning algorithm, under Algorithm 3. Value for variable in calibration is set under camera's initial position e.g. or is searched through orientation, and choice of camera influences output algorithms.

Outline Subject Two prime Setting of Deployment, implementation specialization.

- Encoded observable information etraction from information technology system suchn as HIS, LIS, PACS, EMR. Existing monitoring system for utility traces.

- Mass. Virtuality/Reality synchronized for simulation pre-deploy.

- Examples for virtual spaces of observables include financial time series databses (Large cale Cyberspace). Neuro-Image by IC-Scan in higher-dimension. Internet of Things, all traversable bby method of traversal in physical space by translation Brain network topology representation in structure and dynamics allows for a learners motion, or traversal, over the brain, in pursuit, e.g. of an appropriate Network hyperparametrization (as Networks are cognition inspired). Motion on a guided path would e.g. occur by deep reinforcement learning. (Thus have we foundna possiblity to learn all of a neurak networks parameters)

- Domain specifity: Medicine. Databases intrade and over IoT

Hardware. Drones, data-mills, data collectors, Kardashec Type III data satellties in deep space is no strangeness. If Grok is a Neural netowrk, then can we star-link it with other DL-systems to obtain a DL-Starsystem that owes its basis to the functional archDLStarSystem of the fifth order. All data that but is captured by the StarSystem becomes part of it. One needs data-collector that assume the form of mobile phones or network towers as one would know them from the provsion of internet. Data-mills. Science-function books lay the optimal path through whih to seek for fictitious structures of a scientific nature.

Outline Subject Three Observation, conceptuaization. Navigation. Observation, conceptualization. Setting of deployment, implementation specialization, Setting of deployment.

- Obstacle detection and circumvention motion through space of observables is computable and simulable in framework of Deep Reinforcement Learning. Deep Sea deployment DSMV Miner reconstructible in traversal over every other platform given the required sensors (In DSMV: δ_{tar}, δ_{OLS}, i_{RI}, $\dot{\omega}_b$) inlcuding distance to obstacle. DSMV target path Navigation COntrol Learnable for DSv realization or Land Cramer, Mars Roer, realiation.

- Path-planning in obstacle circumvention feasible by incorporation of setting specific obstacle type in pseudospectral control for unknown ocean curents generalizable to air currents (DOmain-specificty: MODERN PHYSICS)

- Nacigation control manual construction potential starting point in path-planning for underactuated vehicle or roamer by sensor-induced state-representation. Restricted to vision captured by installed visual system (Complication in state-representation must correspond to complecity in area of deployment)

Outline Subject Four Observation, Navigation through observables. Obervation, conceptualization.

- Conv2D Network construction and Chip placement with Deep Reinforcement Learning are one-to-one translatable.

- For non-vision Computation, similar techniques of construction without necessity of feature identification, one-to-one translatability DRL-construction holds.

Navigation through Observables. (Assumes number of observables > 1)

- Necesity of navigation through obervables

- Assumes excessive abundance of observables for preconceptulzaition by observation of one vicinity

- Autonomous vehicle localiation and map-mathcing serve to bear ints towards heuristic boundaries for nacigation towards, or over, preconceptualization. (Momentary example is grounded in extent of conceptualization pre-aperture, and differences therein over contiguous points in space of observables)

- Higher-orbit or deep-sea Ocean/space-craft Navigation policy towards guided exploration or faring towards body, grounded in X-Ray Pulsar Capture may bear hints towards dynamics of observer over spce of obesrvables, the latter yielding a similar extent of mytery and encoding after observation by cconceptulzation

Speculiative Anveunues of Pursuit: Network Design initialization

- Mutiple observer, multi-agent RL over setting of deployment and simukatd setting mirages for aplification of subsets of setting features.
- Globe Maps, Mol Struct COmputational Chip, can all serve as setings, or alyers in simulablemirage of greaeter setting from where these layers can be extracted.
- Redults in Deep Generative Networks will serve as a basis for observing beyond the plain
- Observation Encoding/deconding is not necessry for functionality but is necessary for generation correctness.
- In every Multi-agent deployment can a central coordinator steer over the above points
- Simple setting will servve as foundations for complex setting for correctness verification

Speculative Avenues of Pursuit

Multiple Observers, Mutli-agent RL

- Observer sophistication and device complexity
- Multi-agent, multi-observer, swsrm deployment
- Output type, numeric encoding for alignment with convention
- Continuous observer interpreter re-adjustment
- Deployment in settings of sure exptected dunctionality
- Generalization to new settings to be ancored in deployment to settings of sure expected functionality
- Input sensor interpreter other than Convolutional Neural Network: DLCosmos

  Instaneous setting transformation upon output retrieval.
- Multi-agent, multi-observer for both input and output with static central cordintor
- Every techonolgical vehicle uneer specialzation to its setting can function as an observer, either autonomous or not.

Readily constructibe Avenues of Pursuit.

- Learners over IoT Netwrks
- Learners over Virtual Space
- All the notes written up to the present
- Evaluation performance implicit in (D)RL and DL losses, Heuristic visual serve as heuristic guidance
- Specfication of objectives,
- Device programming and assebly ready for deplyment
- Monitoring learner activity and performance
- Deployment in virtual setting Unity PhysX on small computer
- Deployment over IoT with or without IoT gadgets
- Multi-agent extension towards first most general implementation
- Deplyment on every usual task in data-analysis
- Continuouos lens re-adjustment over movements
- Addition ot subtraction of complexity in learners, construction, more powerful device can give much more complex learner in the same framework current, whether this is good is tobe obsered from performance monitoring.
- Gains in sum of exising trade-offs byobserving movement-capacity for higher observation capacity
- Lens-construction is seting-dependent and setting-distances are reality defined, or learned
- Written documentation for technical specifcation in lens construction by sequential placemnt of elements
- Acquire necessary deviced, bug computers, consoles, IoT gadgets, robot-parts, especially radio-antenna-satellite.
- User-defined travel for observaion beyond the plain

Stages of development

- Manally crafted virtual space deployment machine in assembly CNN-RNN-DMARL or CNN-RNN-DRL using any text-editor ot code-editor
- Unity prepared virtual space deployed machine in assembly CNN-RNN-DRL with camera-agent role taken by built-in Unity Camera's by installing code-editor agent with Unity launcher. Evolve grom simple environments to complex environment, e.g. 2D $\rightarrow$ 3D. Account for the possibility of speculative improvements implementable with Unity, a popular engine for starters.

- Explore different simulation engines, further than MuJoCo's Unity Plug-in or the PhysX-Unity SubEngine for deployment open-world simulations of compex and detailed environments for great information capture, and much space for potential low loss processing, i.e. potential comovement. Improve machines if inmprovement found by Engine's capacity of monitored results, or new inspiration.

- Explore machine-deployment in the real-world, programmed according to machine deployment in the virtual world.

Feature-collection and -identification

- State-specification, Pattern happens by preconception of environment of deplyment to be automated by [Walker-DRL]

- Feature-collection depends on device of observation, thermolens given spectral features that might relate to $\mathcal{D}_f$, and thermolens might be captures by HDR. Feature-collection by Vision-CNN captures many features that are likely entangled with $\mathcal{D}_f$, captured only by an vision-lens

- Feature-collection depends on device of observation. Feature-collection is possible via indicator measurement, as through pH, or a level of chemical-matter extraction as measured by pH, Again, both the latter phenomena may be captures by a Vision-lens. Feature-collection in sound may be processed by CNN by cound-representation in audiogram. Other ways of incorporation towards state s are to be considered for reasons of efficiency. This form of data might rarely be useful.

The problem reads as.

$$[\text{CrossDL}]_{\mathcal{D}_f}(\mathcal{D}_{\text{data}}) \overset{L}{\mapsto} (L_{(i_1,\ldots,i_n)})_{\text{CombPer}(i_1,\ldots,i_n)},$$

for momentary onventient simplicity, consider the problem

$$[\text{CrossDL}]_{\mathcal{D}_f}(\mathcal{D}_{\text{data}}) \overset{L}{\mapsto} (L_{ij})_{\text{CombPer}(ij\in\{1,\ldots n\})}.$$

The function L is defined and results in $(L_{ij})_{ij\in\{1,\ldots,n\}}$ with magnitudes depending on input $[\text{CrossDL}]_{\mathcal{D}_f}(\mathcal{D}_{\text{data}})$. The component $[\text{CrossDL}]_{\mathcal{D}_f}$ depends on finding good representations of $\mathcal{D}_f$. Specification of $\mathcal{D}_f$ is essential and predicatiability, or computability, or lernability depends on this representation.

Pinpointing the problem of, or rather, problems in existing techniques in machine learning Consider $\mathcal{D}_f$, residing in an environment about which one mgiht some preconception, but little notion of theory towrds a represetable property of interest. A first problem is to find the representation, and this might in turn occur by a first application of [Cross-DL] $\leftrightarrows$ [Walker-DRL] equipped only with Vision in an area near $\mathcal{D}_f$, either in non-virtual sense or virtual sense. Afterwards, equipped with representations of $\mathcal{D}_f$, one may re-deply the device for learnability of $\mathcal{D}_f$. Stage 1 deplotment occurs by observation of the phenomenon in a sense that need not yield more clarification than initial encounter, i.e. first contact, towards $\mathcal{D}_f$, that at this

point might knnown by any data-type, e.g. that which made the researcher declare initial encounter by vision. If the representation resulting from the initial Vision enounter succeeds in representing the property of interest, the encounter encoded by $\mathcal{D}_f$ may be used for stage 2 deployment, in which case Stage 1 would result in trhe representation $\mathcal{D}_f$ for encounter $\mathcal{D}_f$

Learning may occur online or offline, and the assumption of data availability is essential. One has some preconception of the relation between target and and relevant features, and therefore features have been selected beforehand. Any preconception may occur on the basis of premonition or estblished theory. Learning the environment happens thereafter on the basis of the latter isolated consideration.

Representing any target or feature, or environment. Many problems in onventional Machine Learning seek to learn for predictability, or explainability. there are phenomena that, however, may barelybe obersable, or envodings that may cause information loss to great extent. One may not observe much latency in a defined numeric feature that does not in te least represent $\mathcal{D}_f$, the ecounter of interest. To learn an enironment is enable accurate generation of the extensions of the environment. In the broadest sense that an environment may reveal itself. The matter of feature-selection occurs itself by learning, and so does the matter of feature-identification occur itelf by learning (from vision lens encoded input). The same would hold for any target, whence more information than mere binary encoding represents an encounter of interest.

Scale It is reasonable to assume that data-entanglement with $\mathcal{D}_f$ occurs near $\mathcal{D}_f$ and while this is of interest for small scale concentional learning, one may be interested in relations that exists over large distances in space, either temporal or physical, or virtual. One far-fetched device than would and capture such an entanglement is a satellite equipped with [Walker-DRL] traversing reality by orbiting earth for mining with [CrossDL] for relations tat attract [Walker-DRL]. The idea, again is to specify an inquiry from which some representable interest follows to be encoded in an alternative reality to which we have access, so that the repressentable target may be reconstructed adn extented. The target may be the means by which first encounter has occurred and may entail an entire physical environment.

The device, be it an orbital satllite or a drone that explores dataspace by traversing planetary orbit and performing areal rend-a-vouz respectively. As one would seek preconception on a target, would the drone traverse an area or a zone to mine for data-entanglements. The problem resides mainly in feture-idetification from e.g. Vision-by-lens, and to define the States in $\mathcal{S}$ so as to optimize environment generation. In summary, th device is the learner as a whole, a completion. We cosntruct and configure the device by specifying the objective, the target. Te device subsequently learns, observer, learns, observes, ..., until reached terminal criterion.

Architectural Design Detail

- [Cross-DL] = [CrossCNNRNN]
- [WalkerDRL] requires care in definition space $\mathcal{S}$, framework POMDP possible
- Test Architecture by deployment

- Follow AlphaChip Architecture for [Walker-DRL] construction.

Vision-CNN Architecture (hyperparameter identification)

- ($\mathcal{D}$-dependent) obs.-CNN$_{\mathcal{D}_f}$ (1 convolutional layer)
- $\mathbf{n}$ (Iteration count application convolutional layers)
- $\mathbf{r}'$ (consideration trade-off training multiple small networks vs. traning one big network)
- λ_n (consideration trade-off training multiple small networks vs. traning one big network)
- $\mathbf{s}$ ((Under optionality of hyperparameter r', subscript $\mathcal{D}$ takes various definitions. In other words, different hyperparameters, inclusion for Vision, different state representation. ([Cross-DL]$^{s}_{\mathcal{D}_f}$)) might be one network, with $\mathcal{D}$-dependent obs-CNN$_{\mathcal{D}_f}$ stacked after the preovous resulting in an optimal network that might the interpretation of an optimal lens.)
- $\mathbf{h}_s$ (definition state historyr, either static past observations over tim-intervals or past sequences of static-observation within one timeframe, corresponds to definition $\mathbf{s}$)

Architectural Design, Short notes Elements are Observation, Interpretation, Exploration, Generation. :Intepreter requires reconfiguration, possible by DRL; Navigation by DRL; Interpretation basis in CNN for interpreter determination $\rightarrow$ follow-up in further CNN in selected interpreter, Genration by application [CrossDL]. First architectural deisgn compeldte $\rightarrow$ software-ready, Improvements follow. Iterate DesignImprovementsFollow $\leftrightarrows$ ArchitecturalImprovementsFollow. Iterate until first device (be it a drone or a satellite) Environment of deployment may vary.

Design of absolute minimality. [Cross-DL] contains only DNN. [Walker-DRL] defined by $\mathcal{S}$: RGB in pixel-value range over $\mathcal{S}$. We denote by $\mathcal{A}$ the device moent over $\mathcal{S}$. We denote by $\mathcal{R}$ the loss. The transitions, or device-specificity, $p(a|s)$, is to be determined by MC simulation. Deploy until break criterion (low loss unlikely by unspecialized [Cross-DL]). Target specification: long measurement by vision, prolonged observation mby measuring device, convetional tool is videocamera. Time interval betwen movements : 4 second to 4 days depending on activity-intensity over $\mathcal{S}$. Local optimum if any action result in negative rewards. Seek for global optimum for set amount of times, after which settle for optimal optimum.

Complication in transitions $p(a|s)$; $\mathcal{S}$ may also be defined by location, or orientation, whence only rewards are derived from Vision observation. Stochasticity in $p(a|s)$ stems from device imperfections os action stochasticity, although determinim in $p(s|a)$ is no obstacle.

Software ready Improvements are possible in definition $s \in \mathcal{S}$, in feature-consideration for $r \in \mathcal{R}$, in device of observation and deployment (dronr better than non-mobile device and $a \in \mathcal{A}$), in target representation, [CrossDL] configurator component

addition, [Walker-DRL] configurator component addition, (configurators for optimization over improvement dimensions), RL optimization, State mixtures in $\mathcal{S}$.

Exhibitability and comptability For compatibility, investigate conventional settings in the current framework. For exhibitability, apply the current framework to settings that extend conventional settings. Compatibility setting examples are to be found on Canvas, usual book exaple on any subject. Exhibitability setting examples; visualize in graphs and deploy virtual agent over graph for any sought after deduction.

Implementation and Visualization Mathematical Software ans software are near immediate. Softwre without desing is mere computation without aim. Mathematics is implied in all written works (see references for details). Software is implied in all written works (see references for details). The design that the written works contain lie at the core. Any convolutional language bears mountain beneath visible tip. There are constraints to which one must adhere. The written works dictate thed direction, code swiftly follow.

Readily achievable objective Learners achieve performance objectives deducible from heuristic visual. Net objectives and devices following from old construction of old devices.

Architectural design Elements include Obseervation, Interpretation, Naigation, Generation.

- For the moment verifications for deplotments are due to simuatlion generation performance
- In oprtmal lens construction and generation write algorithm with AlphaChip Policy network architecture (Capturablew ith RNN Architecture for comparison)
- sift through literature on Navigation, through space of observables for preconceptualization, Leaner construction
- Siftt through liuterauter forobservation learner onstruction, concepetualization learner construction

Vision CNN-Encoder Construction Architecture Construction by greedy features-identification selection is freatestsinplcity. Feature-identification by CNN-Encoding with gredual additional of cnoclutoinal layers of dimensiopns $C_i \times C_o \times K \times K$ where K is the kernel size for specificty in feature identification, C_i is the input channels-0zie (dependson Vision captutere viua depth/color-encodings etc.), C_o is the out-channel size (sets the count of features to be considered in loss-determination and generation). Up-/Down-spampling by NN pooling capturable through specification K and C_o. Selection K, C_i, C_o by input-type. One selection procedure of an algorithmic nature reads as follows.

Capture Vision Input for setting network initialization; input $\leftarrow$ Vision input. BP computation for Loss in Conv2d with single layer of dimensions $C_i \times C_o \times K \times K$ with C_i =input-channel dimension, K=kernel-size=m, $C_o = n$, for

m, n in `range(kernel_limit)`, `range(feature_count_limit)`. Output specification $C_i \times C_o \times K \times K$ for first layer in ConvNetwork (by Loss minimization). Subsequent layers, setting Conv2D Network and consideration next layer, Algorithms for new vision input, Output dimensions (C_o, K for next Conv2D layer.) Choice of Convolutional Neural Network by Reinforcement Learning, Policy Netowrk Architecture, Alpha-Chip. Setting: Conv2D Network Construction and Chip-placement are one-to-one translatable.

Exhibitability and comptability. For today, put al extentsions in every sense on paper, envision big real evice that would run the program, envision every big sci-fi technology that would function as an embodiment of the program. Project inn its echibition may very well be the form that this project yields/ It is just a matter of avenue of application and observer complexity.

An observer interpreter drives the program, Thus seek for many interpreters. Envision interpreter for every setting over which retrieval retrieval might be desired, or needed for furtherment. Data nac take many forms, thus would a setting-spotting Netork be better than intepretation after encoding? What other ways of traveling over a setting exists? A reinforcement learner has i its core built-in the policy, equippable by an observer. Would observation over a setting in its competion be necessary in some situations due to a travelers restrivtion. How would such a activity occur? World are big indeed.

Lens construction occuts by layer-after-layer assembly of a convoluational neural network according to a trained reinforcement learning agent thast steers the typw ans specification and depth of each layer and the network respectively. ANd this has shown effective in the chip-placement analogy. A similar demonstration would have to show the same effectivity for lends construction. What oter ways of observing might there exist? Surely might every network be constructed accoding to a learners dictation. However, how else might an optimel lens be chosen? To be equipped with many alternatives beforehand gives clearance over the flow through which to seek for solutiona that truly work. For code swiftly follows design.

The nature of observation and linking lies at the heart of the programs and their designs, and there are many technologies existant for deducig the one from the other. Sencing rockets to mars is one way to become familiar with the planets, To deploy sub-marines is one way with planetary depths. Another way is to seek for theory and to observe the plain a bit more sharply, and thereby deduce all the rest. Everything conected with everything is an idea not void, and so does the matter of asking the right questions and answering every questions that one might bring to rise be a mere matter of seeking the right notions and deducing thereafter. This is is the idea of preconceptualization. An observer is an eye, interpreter, i.e. conceptualizer and subsequent preconceptualizer as one completion. There exist many implementations for conceptualizers.

Data-collection complication. Feature-collection, Pattern-matching, feature-identification. Data point-entanglemennt: Observation and entanglement identification. Feature-collection and identification. Vision Observer.

$$
\begin{aligned}
\mathcal{D} &\overset{\text{obs.-CNN}}{\rightarrow} (\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_k) \\
&\overset{\text{obs.-CNN}}{\rightarrow} (\mathcal{D}_{11}, \ldots, \mathcal{D}_{1k}, \ldots, \mathcal{D}_{k1}, \ldots, \mathcal{D}_{kk}) \\
&\overset{\text{cross-concat}}{\rightarrow} \text{CombPer}(\mathcal{D}_{11}, \ldots, \mathcal{D}_{1k}, \ldots, \mathcal{D}_{k1}, \ldots, \mathcal{D}_{kk})
\end{aligned}
$$

Vision-entanglemnt: Choose $\mathcal{D}_m$ with $m := \text{argmin}_{ij}\mathcal{D}_{\text{data}}$. CNN-applications: Vision, sound (Aurdiospectrum, Vision), Sense (Thermolens, InfraRed Lens).

Feature-collection 1. Multi-channel CNN in sequential application (Example procedure: Apply Multi-agent CNN with Data-point entanglement treshold reached)
Feature-collection 2. Feature Collection1. Assrance of threshold crossing by movement over environment of deploymeny, encoded in state-space $\mathcal{S}$ for Walker-DRL interpretation.
Feature-collection 3. Feature collection with considered history of feature observaion for capture of feature movement effect on $\mathcal{D}_f$

This is the AI that all want, so read the docs and start architecturing for good implementations.

Conceptualization, pre-conceptualization The deviec would an area and it is known whether optimality is achieved by maximization or minimization or the loss-configuration to define rewards and subsequently train the device that is to perform orbital rend-a-vouz. The orbitsl rend-a-vouz ocurs continuously and is not stopped, The device keeps on training and subsequently perofrms extentensice searcher to become so acquinted with environment as to inger one world from the other, in a manner dictated by the following of policies or a policy optimized towards the following of a route that starts at the accesible world and ends at the world of inquiry. There are reinforcement learning algorithms for this.

Conceptualizationin its many implementations would gain improvement upon completed simulation and attained elements of defect. The RL-lens-sharpener-route of conceptualization, and tweaking its specificity may give improvement, or a translation of another implementation. The RL-route gives leeway for continuous, or discrete (pre-learned) iterative sharpening, an attribute that is not shared with manual initialization or basic grid search automated within time-intervals.

In the current implement, the lens sharpness is adjusted by a leaner that has learned or is learning the opimal adjustment of lenses, the observers navigation protocol has been pre-learned or is being learned by the exploring observer. This is the first implementation, most general.

Vision-CNN Architecture (Hyperparameter identification)

- **CNN optimization** (hyperparameter in $[\text{CrossDL}]^s_{\mathcal{D}_f}$ for optimization, or training optimall CNN for s) (Examples: definition Loss measure, hyperparameters in optimization algorithm (velocity, momentums))

- **RNN optimization** (Hyperparameters in $[\text{CrossDL}]^s_{\mathcal{D}_f}$ for optimization optimal CNN-RNN for s) (Examples: defintion Loss measure, Hyperparameters in optimization algorithm)

- **r** definition rewards, for example by specification of lss-configuration between current state and subsequent state, one example specification is the difference in Loss)
- **a** (type of movement for search preconception depends on search-space, and device of application. See BipedalWalker-v2, e.g. application by drone similar, or traversal dataspace, settlement LunarLander)
- $\mathbf{p}(\mathbf{s}|\mathbf{a})$ (deterministic mainly stochasticity due to uncertainty in environment of application)
- $\pi_\theta(\mathbf{a}|\mathbf{s})$ **optimization** (Hyperparameters in Learning algorithm for optimization $\pi_\theta(\mathbf{a}|\mathbf{s})$, immediate exploration, dictates on-policy learning for example by proximal policy optimization

One elaboration on CNN-encoding of vision observations. Consider the testing for a learner by reinforcement. State start s_0: observation $\mathcal{D}$. Proceed, e.g., as follows.

$$\begin{aligned}
\mathcal{D} &\overset{\text{obs-CNN}}{\mapsto} (\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_\mathrm{k}) \\
&\overset{\text{obs.-CNN}}{\mapsto} (\mathcal{D}_{11}, \ldots, \mathcal{D}_{1k}, \ldots, \mathcal{D}_{k1}, \ldots, \mathcal{D}_{kk}) \\
&\ldots \\
&\overset{\text{obs.-CNN}}{\mapsto} \text{"CombPer"}((\mathcal{D}_{i_1,\ldots,i_n})_{i_1,\ldots,i_n \in \{1,\ldots n\}}) \\
&\overset{L}{\mapsto} \text{"CombPer"}((L_{i_1,\ldots,i_n})_{i_1,\ldots,i_n \in \{1,\ldots n\}}).
\end{aligned}$$

Note that $\mathcal{D}$ is indexable to an order that is exponential to the amount of applications of obs-CNN to observation $\mathcal{D}$. For a representation of states for, e.g. [NavigationDRL], denote therefore

$$\begin{aligned}
s_0 &= (\mathcal{D}_{m_i})_{i=1}^r \\
s_1 &= (\mathcal{D}_{l_i})_{i=1}^r \\
s_2 &= (\mathcal{D}_{k_i})_{i=1}^r \\
&\ldots \\
\hat{r}_i &= L_{s_i} - L_{s_{i-1}}, \quad i \geq 1 \\
\tilde{r}_i &= -\hat{r}_i, \quad i \geq 1 \\
r_i &= \{\hat{r}_i, \tilde{r}_i\}_{i \geq 1}, \quad i \geq 1 \quad \text{Simplest loss-configuration.}
\end{aligned}$$

For positive r, seek optimal loss-configuration, via, e.g. an agent-configuror (see deployment 1.3 for an example hereof). Actions bring transition between states, practical examples of which can be found in drone-rotation, or movement of a virtual entity such as BipedalWalker-v2). Note that, in this simple setting, actions only allow transition between states that are distanced in index by 1. Transition probabilities $p(a|s)$ defined by noise and imperfection in observer and environment may be approximated by initial environment exploration. Policy $\pi_\theta(a|s)$ to be learned by, e.g., DQN. At terminal, settled state, reconsider all kernels (circularities allowed and encouraged). Indication for completion exploration phase may lie in a detected circularity in random traversal. The above example serves to illustrate, and is indeed most simple.

First Practical Detail. Mathematics, SoftwareEngineering, ComputerScience. Simulation Engines: Game Engines: Unity, UnrealEngine, UbiSoftFramework, Anvil. Engines for simulation with pre-installed setting: Celestia, Unity, NVIDIA PhysX, Ansys Fluent, Advanced Simulation Library (OpenSource). Computers: Cloud: LambdaLabs, etc. Scripts deployable in open-world games or engines of simulatio, mainly for capture of virtual setting interconstruction: Heaviest PC Game-type: RTX-9040 Orion Predator Acer, Vector GPU Desktop/Workstation. Gadgets: IoT STM3224 Discovery kit node, ESP32-Azure IoT Kit, IoT-aided artifacts, oculary-autorotators, engine simulation before robotics, oculary must be very high definition.

Mathematics, SoftwareEngineering, ComputerScience. Simulation Engines: Game Engines: Unity, UnrealEngine, UbiSoftFramework, Anvil. Engines for simulation with pre-installed setting: Celestia, Unity, NVIDIA PhysX, Ansys Fluent, Advanced Simulation Library (OpenSource). Computers: Cloud: LambdaLabs, etc. Scripts deployable in open-world games or engines of simulatio, mainly for capture of virtual setting interconstruction: Heaviest PC Game-type: RTX-9040 Orion Predator Acer, Vector GPU Desktop/Workstation. Gadgets: IoT STM3224 Discovery kit node, ESP32-Azure IoT Kit, IoT-aided artifacts, oculary-autorotators, engine simulation before robotics, oculary must be very sharp.